

Tutorial 3 - Q4

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Question 4 Find out which of the following ARMA processes are causal stationary and/or invertible, which are neither causal stationary nor invertible.

a) $X_t = 0.3X_{t-1} - 0.4X_{t-2} + 1.3\epsilon_{t-1} + 0.7\epsilon_{t-2} + \epsilon_t,$

b) $X_t = 1.1X_{t-1} - 0.3X_{t-2} + 1.2\epsilon_{t-1} + \epsilon_t,$

c) $X_t = 0.7X_{t-1} + 0.6X_{t-2} - 0.5\epsilon_{t-1} + 0.4\epsilon_{t-2} + \epsilon_t,$

d) $X_t = 0.8X_{t-1} + 0.3X_{t-2} + 0.6\epsilon_{t-1} - 0.5\epsilon_{t-2} + \epsilon_t,$

where ϵ_t are i.i.d. $N(0, 1)$ random variables.

Follow the following Theorem!

Theorem 5.2 Assume that $\phi(z)$ and $\psi(z)$ have no common factors. Then the ARMA(p, q) process is

1. causal (stationary) iff all roots of $\phi(z)$ lie outside the unit circle,
2. invertible iff all roots of $\psi(z)$ lie outside the unit circle.
3. causal (stationary) and invertible iff all roots of $\phi(z)$ and $\psi(z)$ lie outside the unit circle.

The causal stationarity and invertibility conditions of an ARMA model do not depend on each other,

Characteristic Polynomials:

AR(p)	MA(q)
$\begin{aligned}\phi(z) &= 1 - \phi_1 z - \dots - \phi_p z^p \\ &= 1 - \left(\sum_{i=1}^p \phi_i z^i \right)\end{aligned}$	$\begin{aligned}\psi(z) &= 1 + \psi_1 z + \dots + \psi_q z^q \\ &= 1 + \sum_{i=1}^q \psi_i z^i\end{aligned}$

(a) $X_t = \underbrace{0.3}_{\phi_1} X_{t-1} - \underbrace{0.4}_{\phi_2} X_{t-2} + \underbrace{1.3}_{\psi_1} \epsilon_{t-1} + \underbrace{0.7}_{\psi_2} \epsilon_{t-2} + \epsilon_t,$

ϕ_1 ϕ_2 ψ_1 ψ_2

For Causal Stationary:

$$(1) \quad \phi_1 + \phi_2 = 0.3 - 0.4 = -0.1 < 1$$

$$(2) \quad \phi_2 - \phi_1 = -0.4 - 0.3 = -0.7 < 1$$

$$(3) \quad -1 < (\phi_2 = -0.4) < 1$$

For Invertible:

$$(1) \quad \psi_1 + \psi_2 = 1.3 + 0.7 = 2 > -1$$

$$(2) \quad \psi_1 - \psi_2 = 1.3 - 0.7 = 0.6 < 1$$

$$(3) \quad -1 < (\psi_2 = 0.7) < 1$$

$\therefore X_t$ is causal stationary and invertible

$$\textcircled{b} \quad X_t = \underbrace{1.1}_{\phi_1} X_{t-1} - \underbrace{0.3}_{\phi_2} X_{t-2} + \underbrace{1.2}_{\phi_1} \epsilon_{t-1} + \epsilon_t,$$

For Causal Stationary:

$$\textcircled{1} \quad \phi_1 + \phi_2 = 1.1 - 0.3 = 0.8 < 1$$

$$\textcircled{2} \quad \phi_2 - \phi_1 = -0.3 - 1.1 = -1.4 < 1$$

$$\textcircled{3} \quad -1 < (\phi_2 = -0.3) < 1$$

For Invertible:

$$|\phi_1| \neq 1$$

$\therefore X_t$ is causal stationary but NOT invertible

$$\textcircled{c} X_t = 0.7X_{t-1} + 0.6X_{t-2} - 0.5\epsilon_{t-1} + 0.4\epsilon_{t-2} + \epsilon_t,$$

For Causal Stationary:

$$\textcircled{1} \phi_1 + \phi_2 = 0.7 + 0.6 = 1.3 \neq 1$$

For Invertible:

$$\textcircled{1} \psi_1 + \psi_2 = -0.5 + 0.4 = -0.1 > -1$$

$$\textcircled{2} \psi_1 - \psi_2 = -0.5 - 0.4 = -0.9 < 1$$

$$\textcircled{3} -1 < (\psi_2 = 0.4) < 1$$

$\therefore X_t$ is NOT stationary but is
invertible

$$\textcircled{d} X_t = 0.8X_{t-1} + 0.3X_{t-2} + 0.6\epsilon_{t-1} - 0.5\epsilon_{t-2} + \epsilon_t,$$

For Causal Stationary:

$$\textcircled{1} \phi_1 + \phi_2 = 0.8 + 0.3 = 1.1 \neq 1$$

For Invertible:

$$\textcircled{1} \psi_1 + \psi_2 = 0.6 - 0.5 = 0.1 > -1$$

$$\textcircled{2} \psi_1 - \psi_2 = 0.6 + 0.5 = 1.1 \neq 1$$

$\therefore X_t$ is NOT causal stationary and
is NOT invertible
